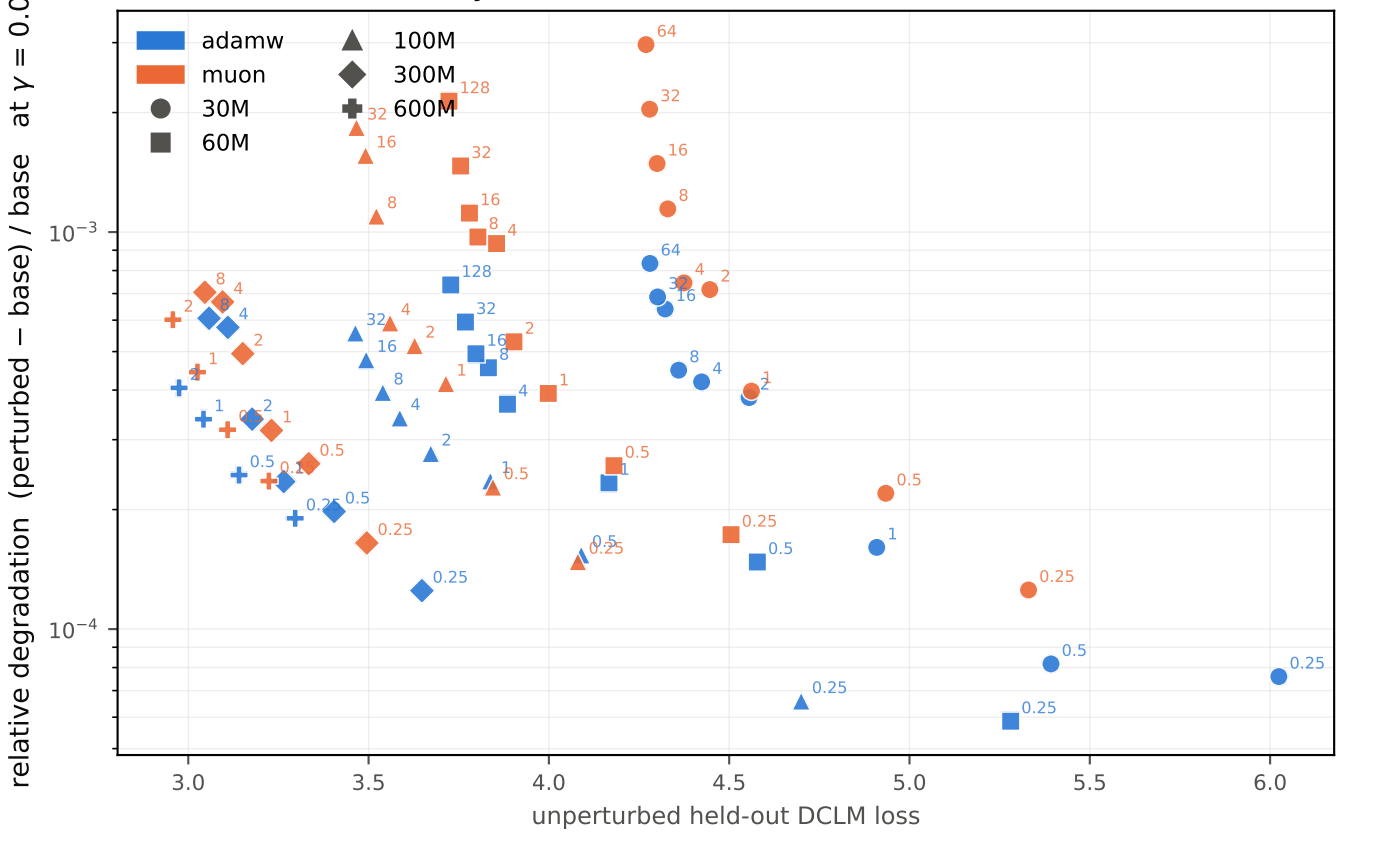
Relative degradation under Gaussian weight perturbation (γ = 0.01) vs model quality
every tuned base; small label = chinchilla
adamw
muon
30M
60M
100M
300M
600M
relative degradation (perturbed − base) / base at γ = 0.01
unperturbed held-out DCLM loss
10^−3
10^−4
3.0
3.5
4.0
4.5
5.0
5.5
6.0